

What the work has established

REPORT 1 / RESULTS OVERVIEW

Scope: the repository snapshot reviewed in this conversation, commit 170fad3, dated 13 September 2026. Findings below are reported measurements, not experiments rerun for this report.

Main finding. The distilled ePC model preserves the BP model very closely. It has not demonstrated an overall advantage as a host for the current cap. Using ePC errors to guide cap learning gives a small paraphrase benefit with substantially poorer acquisition and higher computation.

The system being studied

The base is GPT-2 small: 12 transformer blocks, width 768. A predictive-coding student starts from the BP teacher's weights and is distilled on about **50 million tokens**. The underlying architecture stays the same; explicit error variables and an iterative error solver provide an additional way to obtain learning signals. The cap is a separate memory that changes internal activations while the base weights remain frozen during editing. [R1, R2]

Conversion took about **12.2 local GPU-hours**. The relative weight change was only **0.045%**. This is a faithful conversion, not evidence of a more capable language model. [R1]

How the distilled model is assessed

Assessment	What the results say
Teacher fidelity	Top-choice next tokens agree 99.64% of the time; mean KL divergence is about 0.000029 nats/token.
Ordinary language prediction	Actual next-token accuracy is about 35.8% for both models. Average prediction losses are also almost identical.
Internal features	Feature diversity and part-of-speech probe results are very similar. No clear representational advantage is established.
Correction behavior	Gradient localization and the measured effects of activation writes are broadly similar across the two bases.
ePC error adjustment	Eight-step errors point close to BP correction directions, especially at the last site, but do not meet the convergence criterion.

Interpretation: 99.64% is agreement with the teacher, not accuracy on real text. The current prediction audit uses 199,680 positions; an earlier, larger audit is documented separately. [R3, R4]

What happens when the cap learns

The controlled comparison separates a change of base weights from a change of learning signal.

BP + gradient: original teacher with ordinary gradient-guided cap updates. **ePC + gradient:** distilled base with the same update rule. **ePC + errors:** distilled base with eight iterations of error adjustment supplying the update direction. All use the C1 cap, which writes at all three sites. [R5, R8]

zsRE: 1,000 sequential factual edits

Measured outcome	BP + gradient	ePC + gradient	ePC + errors
Correct immediately after each edit	99.8%	99.9%	66.2%
Original questions correct at the end	52.4%	52.2%	36.6%
Reworded questions correct at the end	13.9%	13.7%	15.6%
Unrelated answers preserved	98.5%	96.3%	98.3%
Learning accelerator seconds/run	272.8	267.6	816.6
Learning + evaluation seconds/run	502.6	499.2	1,045.8

Means over 15 runs per condition: three sealed data realizations, each in five orders. Orders from the same realization are not independent data samples. Retention scores are unconditional scores over the evaluated edits; they are not restricted to edits acquired successfully. Timing rows are calculated from saved phase ledgers. [R5, R6]

What the differences mean

Changing only the base is practically neutral on the primary endpoint. Retained paraphrase accuracy changes by -0.2 percentage points. The locality difference is uncertain: its paired interval includes zero.

Changing the learning signal creates a trade-off. ePC errors improve retained paraphrase accuracy by 1.9 points relative to ePC + gradient, with a reported 97.5% interval of +1.2 to +2.5 points. Immediate acquisition falls by 33.6 points and original-question retention also declines. The study's practical success criteria are not met.

The extra computation is substantial. Error-guided learning uses about 3.0 times the BP learning computation; including evaluation, the run costs about 2.1 times as much. These measurements describe this implementation and update budget.

CounterFact: exact recall without paraphrase transfer

On the 300-edit CounterFact streams, all three conditions score 100% immediate and retained original-answer success, but 0% retained paraphrase success. Their nearly exact matching keys prevent useful retrieval on the tested paraphrases. This result cannot discriminate an ePC generalization advantage. [R5]

What the cap actually is

A memory for changing internal activations at three transformer depths.

At each prediction position, the cap forms a normalized key from the hidden activation. It retrieves the nearest stored key within an allowed radius and adds that entry's correction vector to the activation. If nothing qualifies, the correction is zero. Earlier writes can alter the keys used by later banks. [R7]

Component	Implementation
Attachment sites	After blocks 4, 8 and 12; code indices 3, 7 and 11. The last site is before final layer normalization.
Memory entry	A 768-number key, a 768-number additive correction, a matching radius and metadata.
Storage allowance	About 38.5 MB (36.75 MiB). Nominally 2,048 entries per bank in three-bank variants, before extra bookkeeping.
Learning	Generate a gradient or error direction, try bounded update sizes and commit an improving correction. Base weights remain fixed.
Prediction	Read stored entries and apply their corrections. Prediction does not learn new entries or run target-clamped error inference.

The routing variants

Variant	How a new correction is placed
C0	Use only the final bank, which receives the full memory allowance.
C1	Schedule all three banks; divide the allowed write magnitude among them.
C2	Probe a temporary correction at each bank; select the best positive immediate improvement, or abstain.
CR	Choose a bank randomly using probabilities calibrated from development C2 behavior.

Documentation correction: some narrative passages call C1 single-bank. The implementation and S5 runner use all three banks. C2 is a probe-and-choose rule, not a separately trained neural router. [R8]

What the routing comparison found

In the main BP-based zsRE comparison, C2 retained about 10.9% paraphrase success, versus 13.9% for C1 and 10.9% for CR. Thus the measured routing rule did not improve the primary outcome. C0 was run on shorter 300-edit streams and should not be compared directly with the 1,000-edit endpoints. [R9]

What can and cannot be concluded

The work establishes a useful experimental apparatus and a specific set of limitations.

Established: a faithful ePC conversion; a functioning frozen-base editing cap; controlled substrate and credit comparisons; and substantial acquisition, retention, locality and resource measurements. The ePC solver still uses reverse-mode differentiation, so these results do not demonstrate eliminating backward computation. [R2, R5]

Not established: better language modeling, cheaper continual learning, better causal representations in the distilled GPT-2, or an overall ePC cap advantage. The current cap has no independent predictive-coding energy or internal settling process. [R10]

The synthetic grammar shows reproducible error-space differences between mechanism kinds. It uses a separate BP-trained grammar model with an ePC error solver, so it is not a BP-versus-distilled-GPT-2 comparison. Its confirmatory paraphrase analysis also has a documented seed defect. [R4, R9]

Preservation claims are limited by the actual probes: the saved S5 records use 200 locality prompts and 4,064 scored language-model positions per run. Those are useful checks, not a broad guarantee. GRACE did not qualify for the planned comparison and was not run as a valid confirmatory baseline. [R6, R9]

Short development ablations show a retention gap even when all tested memory ceilings are nonbinding. The grammar's key distances also poorly separate paraphrases from unrelated inputs. These findings motivate improving retrieval and correction expressiveness, alongside a distillation objective that explicitly rewards editability. They do not prove that a particular redesign will succeed. [R9]

Sources and reproducibility

Links below are pinned to the reviewed commit **170fad3ac538ea0df90d0993aa44e2af71628ec0**. Historical reports contain stale labels; this overview uses saved measurements and implementation where they differ. No model training was performed to prepare these PDFs.

R1 Regenerated checkpoint and relative weight change. [docs/tasks/REG-03.md](#).

R2 Error variables, solver, distillation and computation. [docs/epc_energy.md](#).

R3 Current fidelity and language-prediction audit. [results/S1/P1_epc.json](#).

R4 Representation, correction and grammar assessments. [results/S1/report.md](#).

R5 Controlled BP/ePC outcomes and paired analysis. [results/S5/report.md](#).

R6 Evaluation setup; saved results are under results/S4 and results/S5, experiment frozen-confirmatory-v2-84126123. [src/pccap/harness/stage_s5.py](#).

R7 Read/write implementation; bank.py and memory.py specify retrieval and capacity. [src/pccap/cap/cap.py](#).

R8 C1 rule; measured.py specifies C2. [src/pccap/routers/full.py](#).

R9 Integrated results, ablations and deviations. [docs/report.md](#).

R10 Purpose and limits of cap v0. [docs/pc_cap_month_plan_summary.md](#).